

Advanced Cryptography

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Shamir Secret Sharing

ROADMAP

Conceptual View

Secret sharing

Shamir Secret Sharing

Polynomial interpolation

Lagrange Interpolation

Shamir's secret sharing scheme

Proving security of Shamir secret sharing

SECRET SHARING: FORMALLY

Secret Sharing Scheme

Given a message space $\mathcal{M}$, a threshold value $t \in \mathbb{N}$, and a number of shares $n \in \mathbb{N}$, with $t < n$.

A (t, n) **threshold secret sharing scheme** consists of two algorithms:

Share: takes a message $m \in \mathcal{M}$ as input, and outputs an n -tuple of randomly selected shares $\mathbf{s} = (s_1, \dots, s_n)$.

Reconstruct: takes a subset of t or more shares as input, and produces a message m' as output.

Notice the word “randomly” in **Share**: it implies that this is a *probabilistic algorithm*.

Notice that we have, so far, only looked at (n, n) schemes: going towards (t, n) is not straightforward.

SECRET SHARING TOWARDS (t, n)

Mathematics provides several tricks to get (t, n) schemes.

What we are looking for are mathematical problems, where we can “reconstruct” something based on at least t variables.

For instance, consider the Chinese Remainder Theorem (you should be familiar with it from RSA?): it says that given t values modulo t primes, we can reconstruct the unique value modulo the product of the t primes.

For instance, consider intersecting hyperplanes: to define a point in t dimensional space, you need t hyperplanes.

Both of these ideas were used to define secret sharing schemes, neither was too popular in practice.

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POLYNOMIALS

A **polynomial** is a mathematical expression where (finitely many) variables and coefficients are added, multiplied, subtracted and raised to a non-negative power. For instance a polynomial in the variable x is written as

$$f(x) = a_d x^d + a_{d-1} x^{d-1} + \dots a_1 x + a_0.$$

The highest non-zero term determines the **degree** of a polynomial, e.g. $f(x) = x^2 - 1$ has degree 2.

We can evaluate $f(x) = x^2 - 1$ over the real numbers by selecting $x = 2 : f(2) = 3$.

POLYNOMIALS OVER $\mathbb{Z}_p$

We can evaluate $f(x) = x^2 - 1$ over $\mathbb{Z}_3$ in which case $f(2) \equiv 0$.

When working modulo p , because $x^p \equiv x \pmod{p}$, we always assume that exponents are strictly smaller than p (otherwise we reduce the exponent modulo $p - 1$); with this we ensure that there is a unique function expressing a polynomial.

The set of polynomials over $\mathbb{Z}_p$ is a commutative ring:

- ▶ if we take two polynomials over $\mathbb{Z}_p$ then their sum exist,
- ▶ their difference exists, and
- ▶ we can multiply two polynomials,
- ▶ and there is a division with remainder.

For instance dividing $6x^4 + 8x + 1$ by $2x^2 + 4$ over $\mathbb{Z}_{11}$ gives $(3x^2 + 5)(2x^2 + 4) + (8x + 3)$.

IMPORTANT PROPERTIES OF POLYNOMIALS

Property 1. A non-zero polynomial of degree d has at most d roots.

Property 2. Given $d + 1$ distinct pairs $(x_1, y_1), \dots, (x_{d+1}, y_{d+1})$, there is a unique polynomial $f(x)$ such that $f(x_i) = y_i$ for $1 \leq i \leq d + 1$.

Number of polynomials of degree d over $\mathbb{Z}_p$: A degree d polynomial has $d + 1$ coefficients, there are p choices for each coefficient, and thus there are p^{d+1} polynomials.

Suppose we have d points given: these will enable us to fix all but one coefficient, implying that there are p polynomials.

If $d - k$ pairs of points are given, then there are p^{d+1-k} polynomials.

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RECONSTRUCT A POLYNOMIAL GIVEN SEVERAL POINTS, TOY EXAMPLE

We have the pairs $(1, f(1) = 2)$, $(2, f(2) = 0)$, $(6, f(6) = 7)$, $(8, f(8) = 10)$ in $\mathbb{Z}_{11}$.

There is a unique polynomial of degree 3: $f(x) = a_3x^3 + a_2x^2 + a_1x + a_0$ over $\mathbb{Z}_{11}$ with these four points.

We plug the numbers in, and get four equations in four unknowns:

$$2 = a_3 + a_2 + a_1 + a_0$$

$$0 = 8a_3 + 4a_2 + 2a_1 + a_0$$

$$7 \equiv 7a_3 + 3a_2 + 6a_1 + a_0$$

$$10 \equiv 6a_3 + 8a_1 + a_0$$

The unique degree three polynomial is $f(x) = x^3 + 2x + 10 \pmod{11}$.

LAGRANGE INTERPOLATION

Given $d + 1$ pairs $(x_i, f(x_i))$, construct the so-called Lagrange polynomials

$$\delta_i(x) = \frac{\prod_{j \neq i} (x - x_j)}{\prod_{j \neq i} (x_i - x_j)}.$$

These polynomials have the property that given a pair $(x_i, f(x_i))$, the polynomial $\delta_i(x_i)$ evaluates to 1 and it evaluates to 0 otherwise.

Therefore the polynomial $f(x)$ that we are looking for is given as:

$$f(x) = \sum_{i=1}^{d+1} f(x_i) \delta_i(x).$$

Note that $f(x_i)$ is NOT a function here but a value in $\mathbb{Z}_p$.

EXAMPLE, CONT.

For the first point $(1, f(1) = 2)$, the corresponding Lagrange polynomial is

$$\delta_1(x) = \frac{(x-2)(x-6)(x-8)}{(1-2)(1-6)(1-8)} \equiv (x^3 + 6x^2 + 10x + 3)9^{-1}.$$

The other Lagrange polynomials (i.e. for $(2, f(2) = 0)$, $(6, f(6) = 7)$, $(8, f(8) = 10)$) are:

$$\delta_2(x) = 5(x^3 + 7x^2 + 7x + 7)$$

$$\delta_3(x) = 3(x^3 + 4x + 6)$$

$$\delta_4(x) = 8(x^3 + 2x^2 + 9x - 1)$$

$$\begin{aligned} f(x_1)\delta_1(x) + f(x_2)\delta_2(x) + f(x_3)\delta_3(x) + f(x_4)\delta_4(x) = \\ -(x^3 + 6x^2 + 10x + 3) + 0 - (x^3 + 4x + 6) + 3(x^3 + 2x^2 + 9x - 1) = x^3 + 2x + 10 \end{aligned}$$

ANOTHER EXAMPLE, OVER TO YOU!

Find the polynomial that passes through the points $(1, 1)$, $(2, 3)$, and $(4, 9)$ whereby the coordinates are in $\mathbb{Z}_{13}$.

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SHAMIR'S SCHEME: INFORMALLY

A secret $m \in \mathbb{Z}_p$ is to be shared with a threshold of t .

We choose a secret degree $(t - 1)$ polynomial f , that satisfies $f(0) \equiv m \pmod{p}$.

This is another way of saying that the coefficient a_0 in our notation is exactly $m \pmod{p}$.

All other coefficients are chosen randomly from $\mathbb{Z}_p$.

The i -th user receives the point $(i, f(i) \pmod{p})$ (and $i > 0$).

It is of course possible to give a user more than one share/point if so desired in a concrete situation.

SHAMIR'S SCHEME: SHARE AND RECONSTRUCT

```
1  def Stn.Share(m):  
2       $a_1, \dots, a_{t-1} \xleftarrow{\$} \mathbb{Z}_p$   
3       $f(x) = m + \sum_{j=1}^{t-1} a_j x^j$   
4      for i=1 to n:  
5           $s_i = (i, f(i) \pmod p)$   
6      return  $\mathbf{s} = (s_1, \dots, s_n)$ 
```

Listing: Shamir (t, n) secret sharing: Share function

```
1  def Stn.Reconstruct( $\{s_i | i \in \mathcal{U}\}$ ):  
2      If  $|\mathcal{U}| \geq t$ :  
3          Derive unique degree  $t-1$  polynomial  $f(x)$   
4          return  $f(0)$   
5      else  
6          return  $\perp$ 
```

Listing: Shamir (t, n) secret sharing: Reconstruct function

EXAMPLE

There are five parties in total and we want a threshold $t = 3$ (i.e. at least three parties must come together to reveal the secret).

We (= holder of the secret, aka “dealer”) therefore must construct a polynomial of degree 2 (this will have 3 coefficients), so $f(x) = m + a_1x + a_2x^2$.

Suppose that $m = 7$, and we randomly sample $a_1 = 4$ and $a_2 = 4$ in $\mathbb{Z}_{11}$.

Now we can create five shares: $(1, 1), (2, 8), (3, 6), (4, 6), (5, 8)$ (the y -coordinate is the share).

EXAMPLE, CONT.

Suppose that parties 3, 4, 5 come together. They can reconstruct the secret by first computing the Lagrange polynomials:

$$\delta_3(x) = 6x^2 + 12x + 54$$

$$\delta_4(x) = 10x^2 + 8x + 7$$

$$\delta_5(x) = 6x^2 + 2x + 6$$

Next by deriving the secret polynomial $f(x) = f(3)\delta_3(x) + f(4)\delta_4(x) + f(5)\delta_5(x)$.

And then evaluating it at zero $m = f(0)$.

Notice that one could in fact publish a “reconstruction vector” for the set of users 3, 4, 5 so that they can evaluate $f(0)$ without needing to explicitly construct Lagrange polynomials.

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SECURITY ARGUMENT (INFORMALLY)

m is an element of $\mathbb{Z}_p$, that was chosen with some probability.

All the a_i where chosen independently and randomly from $\mathbb{Z}_p$.

Consider the “worst case” where all but a single share fall into the hands of an adversary;
say the t -th share is still unknown:

$$m - \textcolor{red}{y}_t \delta_t(0) = \sum_{i=1}^{t-1} y_i \delta_i(0).$$

$\delta_t(0) \neq 0$ (per construction).

$$m - \textcolor{red}{y}_t \cdot r = s$$

If y_t is uniformly distributed, then this is like a one-time pad encryption of m over $\mathbb{Z}_p$.
Knowledge of the “ciphertext” s gives no (additional) information about m .

ARE THE SHARES UNIFORMLY DISTRIBUTED?

Yes: the coefficients a_i of the polynomial are selected uniformly at random; they define exactly one polynomial $f(x)$.

The resulting polynomial $f(x)$ then gives the points/shares $(x_i, f(x_i))$. There is an injective mapping from the coefficients to points/shares.

Given shares we can invert this mapping: a set of d shares (=points) uniquely defines $f(x)$ (aka a set of d coefficients in $\mathbb{Z}_p$). Thus the mapping between shares to coefficients is surjective.

Thus the mapping between the coefficients and shares is a bijection.

A bijection preserves the uniform probability distribution: because the a_i are uniform, the $f(i)$ are uniform too!

SECURITY: MORE FORMAL

According to our security definition, we want to show that an adversary cannot distinguish between the shares of two messages m, m' .

Bear in mind that our adversary may corrupt up to $t - 1$ parties.

This can be done by first using a simulation based argument, i.e. where we show that we cannot distinguish between a real execution of the protocol and an execution in an ideal world.

And then we use the fact that the real and random world are indistinguishable to show that also a Left/Right Oracle are indistinguishable.

REAL AND IDEAL=RANDOM ORACLES

```
1  def Share(m, t,  $\mathcal{U} \subset \{1, \dots, p\}$ ) :  
2      if  $|\mathcal{U}| < t$ :  
3          return  $\perp$   
4       $a_1, \dots, a_{t-1} \xleftarrow{\$} \mathbb{Z}_p$   
5       $f(x) = m + \sum_{j=1}^{t-1} a_j x^j$   
6      for i=1 to n:  
7           $s_i = (i, f(i) \pmod{p})$   
8      return  $\{s_i | i \in \mathcal{U}\}$ 
```

Listing: Real world

```
1  def Share(m, t,  $\mathcal{U} \subset \{1, \dots, p\}$ ) :  
2      if  $|\mathcal{U}| < t$ :  
3          return  $\perp$   
4      for  $i \in \mathcal{U}$ :  
5           $y_i \xleftarrow{\$} \mathbb{Z}_p$   
6           $s_i = (i, y_i)$   
7      return  $\{s_i | i \in \mathcal{U}\}$ 
```

Listing: Ideal world

SEQUENCE OF STEPS IN A FORMAL PROOF ARGUMENT: REAL WORLD

The probability by which a given input $m, t, \mathcal{U}$ produces an output i, y_i depends on the random choice in line 2, and the “restriction” that the input puts on the polynomial.

The polynomial runs through $(0, m)$ as well as the outputs (i, y_i) . We sample $t - 1$ random values from $\mathbb{Z}_p$ in line two, which implies that there are p^{t-1} polynomials possible from that choice in line 2.

We select one of them which must go through the given $|\mathcal{U}| + 1$ points; there are exactly $p^{t-1-|\mathcal{U}|}$ such polynomials.

Thus the overall probability to select a polynomial with $t - 1$ random coefficients that go through some given $|\mathcal{U}|$ points is

$$\frac{p^{t-1-|\mathcal{U}|}}{p^{t-1}} = p^{-|\mathcal{U}|}.$$

SEQUENCE OF STEPS IN A FORMAL PROOF ARGUMENT: IDEAL WORLD

The library corresponding to the ideal world samples $|\mathcal{U}|$ values from $\mathbb{Z}_p$ in line 5. There are $p^{|\mathcal{U}|}$ choices here thus the given set (i, y_i) is sampled with probability $p^{-|\mathcal{U}|}$.

As we placed no conditions/assumptions/restrictions on the input $m, t, \mathcal{U}$, we can now conclude that for all inputs, both libraries return (i, y_i) with the same probability, which is $p^{-|\mathcal{U}|}$.

Therefore the libraries are indistinguishable for an adversary: they have the same interface, and we showed that the probability distribution of their outputs is identical.

TOWARDS INDISTINGUISHABLE SHARES: GIVE IT A TRY

Using the fact that we are secure in a real/random setting enables us to show security in the Left/Right setting.

Start with a call to e.g. the Left oracle.

Notice: the Left Oracle corresponds to the Real Oracle ...